Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?Answer Choices:A. It activates TLR5 on the surface of macrophages.B. It primarily induces the Th1-cell response.C. Peptidoglycan is its major antigen that induces an immune response.D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>The patient's symptoms and clinical presentation are consistent with community-acquired pneumonia. The X-ray shows patchy reticular opacities in the perihilar regions of both lungs, which is commonly seen in Streptococcus pneumoniae infection. Streptococcus pneumoniae is a gram-positive bacterium that has peptidoglycan as a major component of its cell wall. The immune response to this pathogen typically involves the activation of Toll-like receptors (TLR) on the surface of macrophages, particularly TLR2, which recognize the peptidoglycan. The TLR2 activation leads to an immune response involving both the innate and adaptive immune systems. The patient's response to cephalexin suggests that the pathogen is sensitive to this antibiotic, which is effective against Streptococcus pneumoniae. The immune response to Streptococcus pneumoniae is characterized by a Th1-cell response, as the bacteria are primarily extracellular and the response is directed against its cell wall components. Streptococcus pneumoniae does not typically evade the immune response through encapsulation; in fact, its capsule is a target of the immune response.</think><answer>C. Peptidoglycan is its major antigen that induces an immune response.</answer>